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Year 10 Mathematics**Investigation 3, 2015****Topic – Division of Polynomials****Complete in-class Section****Important Information:**

Although the complete in-class component is not worth any marks, it is essential in preparation for the validation component. Knowledge and skills gained will be extended in the validation component. This in-class validation will be completed under test conditions on the day in which this complete in-class component is due. The complete in-class component may be used when completing the validation.

Date out:	Week _____ Date _____	Date Due:	Week _____ Date _____
Take home component weighting:	0% of the year	In-class component weighting:	10 % of the semester

In this investigation you will learn how to use long division to factorise polynomials larger than quadratics. It is essential that you learn and duplicate the methods detailed in this investigation to enable you to correctly factorise the high order polynomials that will be presented later in this investigation and in the validation section.

One method of dividing numbers is as follows. Although many may have been taught other methods at Primary School, it is important that you learn the following method, as we will extend it to use with polynomials.

If I wanted to divide 86 by 3, then I would do the following:

$$3 \overline{) 86}$$

3 goes into 8 twice, so place a 2 above the 8

$$3 \overline{) 86} \quad \begin{array}{r} 2 \\ \hline \end{array}$$

Now multiply 3 by 2 and place below the 8, and then subtract

$$\begin{array}{r} 2 \\ 3 \overline{) 86} \\ \underline{-6} \\ 2 \end{array}$$

Next bring the 6 down from the 86 and put beside the 2

$$\begin{array}{r} 2 \\ 3 \overline{) 86} \\ \underline{-6} \downarrow \\ 26 \end{array}$$

3 goes into 26 8 times, so place an 8 beside the two on the top row

$$\begin{array}{r} 28 \\ 3 \overline{) 86} \\ \underline{-6} \downarrow \\ 26 \end{array}$$

And again 3 times 8 equals 24

$$\begin{array}{r} 28 \text{ r } 2 \\ 3 \overline{) 86} \\ \underline{-6} \downarrow \\ 26 \\ \underline{-24} \\ 2 \end{array}$$

So $86 \div 3 = 28$ remainder 2

Example 1

$$\begin{array}{r} 198 \text{ r } 6 \\ 7 \overline{) 1392} \\ \underline{-7} \downarrow \\ 69 \\ \underline{-63} \downarrow \\ 62 \\ \underline{-56} \\ 6 \end{array}$$

$$1392 \div 7 = 198 \text{ remainder } 6$$

Example 2

$$\begin{array}{r} 406 \text{ r } 4 \\ 5 \overline{) 2034} \\ \underline{-20} \downarrow \downarrow \\ 034 \\ \underline{-30} \\ 4 \end{array}$$

Since 5 doesn't go into 3, place a 0 here and bring down the next number (the 4)

$$2034 \div 5 = 406 \text{ remainder } 4$$

Your turn, using the long division method shown above, divide the following.

1. $223 \div 7$

$$\begin{array}{r} 31 \text{ r } 6 \\ 7 \overline{) 223} \\ \underline{21} \\ 13 \\ \underline{7} \\ 6 \end{array}$$

2. $32 \div 3$

$$\begin{array}{r} 10 \text{ r } 2 \\ 3 \overline{) 32} \\ \underline{3} \\ 02 \end{array}$$

3. $625 \div 2$

$$\begin{array}{r} 312 \text{ r } 1 \\ 2 \overline{) 625} \\ \underline{6} \\ 02 \\ \underline{2} \\ 05 \\ \underline{4} \\ 1 \end{array}$$

4. $5237 \div 5$

$$\begin{array}{r} 1047 \text{ r } 2 \\ 5 \overline{) 5237} \\ \underline{5} \\ 023 \\ \underline{20} \\ 37 \\ \underline{35} \\ 2 \end{array}$$

5. $8702 \div 3$

$$\begin{array}{r} 2900 \text{ r } 2 \\ 3 \overline{) 8702} \\ \underline{6} \\ 27 \\ \underline{27} \\ 002 \end{array}$$

6. $62\,785 \div 4$

$$\begin{array}{r}
 15696 \text{ r } 1 \\
 4 \overline{) 62785} \\
 \underline{4} \\
 22 \\
 \underline{20} \\
 27 \\
 \underline{24} \\
 38 \\
 \underline{36} \\
 25 \\
 \underline{24} \\
 1
 \end{array}$$

We can do the same process with polynomials. Now recall a polynomial is a sum of terms that have a constant and a power of x

e.g. $P(x) = 8x^4 + 3x^3 - 2x^2 - x + 8$

The **degree** of the polynomial is the highest power, so the degree of the example would be 4.

The **leading term** is the term that contains the highest power, so $8x^4$ is the leading term.

We set the division of polynomials up just like we did for numbers.

Let us divide $4x^3 - 9x^2 + 6x - 1$ by $x - 2$

$$x - 2 \overline{) 4x^3 - 9x^2 + 6x - 1}$$

Starting with the leading term, we need to find **what** multiplied by x is equal to $4x^3$. That would be $4x^2$, so place $4x^2$ above the leading term.

$$\begin{array}{r}
 4x^2 \\
 x - 2 \overline{) 4x^3 - 9x^2 + 6x - 1}
 \end{array}$$

Now multiply $x - 2$ by $4x^2$, and subtract.

Careful with the subtraction of negatives $\longrightarrow$

$$\begin{array}{r}
 4x^2 \\
 x - 2 \overline{) 4x^3 - 9x^2 + 6x - 1} \\
 \underline{-(4x^3 - 8x^2)} \\
 x^2
 \end{array}$$

Bring down the "+6x"

$$\begin{array}{r}
 4x^2 \\
 x - 2 \overline{) 4x^3 - 9x^2 + 6x - 1} \\
 \underline{-(4x^3 - 8x^2)} \quad \downarrow \\
 -x^2 + 6x
 \end{array}$$

Next multiply $x - 2$ by " $-x$ " to get the leading term of our answer

$$\begin{array}{r}
 4x^2 - x \\
 x - 2 \overline{) 4x^3 - 9x^2 + 6x - 1} \\
 \underline{-(4x^3 - 8x^2)} \quad \downarrow \\
 -x^2 + 6x \\
 \underline{-(-x^2 + 2x)} \\
 4x - 1
 \end{array}$$

Once again subtract and then bring down the "- 1"

$$\begin{array}{r}
 4x^2 - x \\
 x - 2 \overline{) 4x^3 - 9x^2 + 6x - 1} \\
 \underline{-(4x^3 - 8x^2)} \quad \downarrow \quad \downarrow \\
 -x^2 + 6x \\
 \underline{-(-x^2 + 2x)} \quad \downarrow \\
 4x - 1
 \end{array}$$

Multiply $x - 2$ by 4 to get the new leading term and again subtract.

$$\begin{array}{r}
 4x^2 - x + 4 \text{ r } 7 \\
 x - 2 \overline{) 4x^3 - 9x^2 + 6x - 1} \\
 \underline{-(4x^3 - 8x^2)} \quad \downarrow \quad \downarrow \\
 -x^2 + 6x \\
 \underline{-(-x^2 + 2x)} \quad \downarrow \\
 4x - 1 \\
 \underline{-(4x - 8)} \\
 7
 \end{array}$$

$$\text{So } 4x^3 - 9x^2 + 6x - 1 = (4x^2 - x + 4)(x - 2) + 7$$

$\uparrow$ $\uparrow$ $\uparrow$ $\uparrow$
 Dividend Quotient Divisor Remainder

Each part of our answer has a special name and should be remembered.

Example 3

Divide $3x^4 - 2x^3 + 7x^2 - x + 1$ by $x - 3$

$$\begin{array}{r}
 3x^3 + 7x^2 + 28x + 83 \text{ r } 250 \\
 x - 3 \overline{) 3x^4 - 2x^3 + 7x^2 - x + 1} \\
 \underline{3x^4 - 9x^3} \downarrow \\
 7x^3 + 7x^2 \downarrow \\
 \underline{7x^3 - 21x^2} \downarrow \\
 28x^2 - x \downarrow \\
 \underline{28x^2 - 84x} \downarrow \\
 83x + 1 \downarrow \\
 \underline{83x - 249} \\
 250
 \end{array}$$

So $3x^4 - 2x^3 + 7x^2 - x + 1 = (3x^3 + 7x^2 + 28x + 83)(x - 3) + 250$

Example 4

Divide $3x^5 + 2x^3 + x^2 + 5$ by $x + 1$

Firstly we have no " x^4 " term and no " x " term, so set out as follows

$$x + 1 \overline{) 3x^5 + 0x^4 + 2x^3 + x^2 + 0x + 5}$$

We place terms with coefficients of zero to make sure every power is represented in the dividend.

$$\begin{array}{r}
 3x^4 - 3x^3 + 5x^2 - 4x + 4 \\
 x + 1 \overline{) 3x^5 + 0x^4 + 2x^3 + x^2 + 0x + 5} \\
 \underline{3x^5 + 3x^4} \downarrow \\
 -3x^4 + 2x^3 \downarrow \\
 \underline{-3x^4 - 3x^3} \downarrow \\
 5x^3 + x^2 \downarrow \\
 \underline{5x^3 + 5x^2} \downarrow \\
 -4x^2 + 0x \downarrow \\
 \underline{-4x^2 - 4x} \\
 4x + 5 \\
 \underline{4x + 4} \\
 1
 \end{array}$$

So $3x^5 + 2x^3 + x^2 + 5 = (3x^4 - 3x^3 + 5x^2 - 4x + 4)(x + 1) + 1$

Exercises

7. Divide $x^2 - x - 2$ by $x + 2$

$$\begin{array}{r} x - 3 \\ x + 2 \overline{) x^2 - x - 2} \\ \underline{x^2 + 2x} \\ -3x - 2 \\ \underline{-3x - 6} \\ 4 \end{array}$$

$$(x + 2)(x - 3) + 4$$

8. Divide $2x^3 - 3x^2 - 2x + 1$ by $x + 3$

$$\begin{array}{r} 2x^2 - 9x + 25 \\ x + 3 \overline{) 2x^3 - 3x^2 - 2x + 1} \\ \underline{2x^3 + 6x^2} \\ -9x^2 - 2x \\ \underline{-9x^2 - 27x} \\ 25x + 1 \\ \underline{25x + 75} \\ -74 \end{array}$$

$$(x + 3)(2x^2 - 9x + 25) - 74$$

9. Divide $3x^3 - 3x^2 + x + 2$ by $x + 1$

$$\begin{array}{r}
 3x^2 + 0x + 1 \\
 x+1 \overline{) 3x^3 - 3x^2 + x + 2} \\
 \underline{3x^3 + 3x^2} \\
 0 + x + 2 \\
 \underline{x + 1} \\
 1
 \end{array}$$

$$(x+1)(3x^2+1) + 1$$

10. Divide $2x^3 + 2x + 1$ by $x - 2$ (see example 4)

$$\begin{array}{r}
 2x^2 + 2x + 6 \\
 x-2 \overline{) 2x^3 + 0x^2 + 2x + 1} \\
 \underline{2x^3 - 2x^2} \\
 2x^2 + 2x \\
 \underline{2x^2 - 4x} \\
 6x + 1 \\
 \underline{6x - 12} \\
 13
 \end{array}$$

$$(x-2)(2x^2+2x+6) + 13$$

In formal language division of polynomials is:

For any polynomial $a(x)$ and $b(x)$, where $b(x)$ is not a constant, there exists unique polynomials $q(x)$ and $r(x)$ such that

$$a(x) = q(x) \times b(x) + r(x)$$

Where $r(x)$ is less than the degree of $b(x)$

$a(x)$ is the dividend
 $q(x)$ is the quotient
 $b(x)$ is the divisor, and
 $r(x)$ is the remainder

Finding factors of polynomials

If we divided a number by another and the remainder is zero, then the divisor is a factor of the dividend.

e.g. $20 \div 4 = 5 \text{ r } 0$, so 4 is a factor of 20

The same applies to polynomials. If $r(x) = 0$ then $b(x)$ is a factor of $a(x)$

The Remainder Theorem

When a polynomial $p(x)$ is divided by $x - a$, the remainder is $p(a)$

The Factor Theorem

$p(a) = 0$ if and only if $x - a$ is a factor of the polynomial $p(x)$

Example 5

Check if $x + 1$ is a factor of $x^3 + 6x^2 + 11x + 6$

$$\text{Let } p(x) = x^3 + 6x^2 + 11x + 6$$

Rewrite $x + 1$ in the form $x - a$, $x - -1$ and $a = -1$

Substitute into $p(x)$

$$\begin{aligned} p(-1) &= (-1)^3 + 6(-1)^2 + 11(-1) + 6 \\ &= -1 + 6 - 11 + 6 \\ &= 0 \end{aligned}$$

So $(x + 1)$ is a factor of $x^3 + 6x^2 + 11x + 6$

Example 6

Show that $x - 2$ and $x + 3$ are both factors of $x^4 - 3x^3 - 15x^2 + 19x + 30$

Let $p(x) = x^4 - 3x^3 - 15x^2 + 19x + 30$

With $(x - 2)$, $a = 2$, so

$$\begin{aligned} p(2) &= 2^4 - 3 \times 2^3 - 15 \times 2^2 + 19 \times 2 + 30 \\ &= 16 - 24 - 60 + 38 + 30 \\ &= 0 \end{aligned}$$

Thus $(x - 2)$ is a factor

With $(x + 3)$, re-write as $(x - -3)$ therefore $a = -3$;

$$\begin{aligned} p(-3) &= (-3)^4 - 3 \times (-3)^3 - 15 \times (-3)^2 + 19 \times (-3) + 30 \\ &= 81 + 81 - 135 - 57 + 30 \\ &= 0 \end{aligned}$$

Thus $(x + 3)$ is a factor

Exercises

11. Show that $x + 1$ and $x - 2$ are both factors of $x^2 - x - 2$

$$p(x) = x^2 - x - 2$$

$$(x + 1) = (x - -1) \therefore a = -1$$

$$\begin{aligned} p(-1) &= (-1)^2 - (-1) - 2 \\ &= 1 + 1 - 2 \\ &= 0 \end{aligned}$$

$\therefore x + 1$ is a factor

$$(x - 2) \therefore a = 2$$

$$\begin{aligned} p(2) &= 2^2 - 2 - 2 \\ &= 4 - 4 \\ &= 0 \end{aligned}$$

$\therefore x - 2$ is a factor

12. Show that $x + 1$ and $x - 2$ are both factors of $x^3 - 3x^2 + 4$

$$p(x) = x^3 - 3x^2 + 4$$

$$(x + 1) = (x - -1) \therefore a = -1$$

$$\begin{aligned} p(-1) &= (-1)^3 - 3(-1)^2 + 4 \\ &= -1 - 3 + 4 \\ &= 0 \end{aligned}$$

$\therefore x + 1$ is a factor

$$(x - 2) \therefore a = 2$$

$$\begin{aligned} p(2) &= 2^3 - 3(2)^2 + 4 \\ &= 8 - 12 + 4 \\ &= 0 \end{aligned}$$

$\therefore x - 2$ is a factor

13. Show that $x+3$ and $x-3$ are both factors of $x^4 + x^3 - 15x^2 - 9x + 54$

$$p(x) = x^4 + x^3 - 15x^2 - 9x + 54$$

$$(x+3) = (x - (-3)) \therefore a = -3$$

$$\begin{aligned} p(-3) &= (-3)^4 + (-3)^3 - 15(-3)^2 - 9(-3) + 54 \\ &= 81 - 27 - 135 + 27 + 54 \\ &= 0 \end{aligned}$$

$\therefore (x+3)$ is a factor

$$(x-3) \therefore a = 3$$

$$\begin{aligned} p(3) &= 3^4 + 3^3 - 15 \times 3^2 - 9 \times 3 + 54 \\ &= 81 + 27 - 135 - 27 + 54 \\ &= 0 \end{aligned}$$

$\therefore (x-3)$ is a factor

14. Show that $x+1$ and $x+2$ are both factors of $x^3 - x^2 - 10x - 8$

$$p(x) = x^3 - x^2 - 10x - 8$$

$$(x+1) = (x - (-1)) \therefore a = -1$$

$$\begin{aligned} p(-1) &= (-1)^3 - (-1)^2 - 10(-1) - 8 \\ &= -1 - 1 + 10 - 8 \\ &= 0 \end{aligned}$$

$\therefore x+1$ is a factor

$$(x+2) = (x - (-2)) \therefore a = -2$$

$$\begin{aligned} p(-2) &= (-2)^3 - (-2)^2 - 10(-2) - 8 \\ &= -8 - 4 + 20 - 8 \\ &= 0 \end{aligned}$$

$\therefore x+2$ is a factor

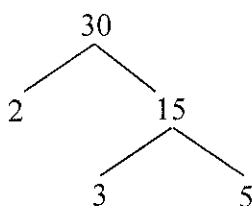
Once we know if a term is a factor, we can use long division to remove the factor from the polynomial. This process is then repeated on the quotient until the polynomial is fully factorized.

Example 7

Factorise $x^4 - 3x^3 - 15x^2 + 19x + 30$

Some things to think about:

- The degree of the polynomial tells you how many sets of brackets to expect at the end of the factorisation. x^4 can only be got by $x \times x \times x \times x$, so 4 brackets.
- The constant term (30 in this case) is obtained by multiplying the numbers in each bracket together, so our 30 is made up by multiplying 4 numbers together. A prime factorisation of 30 will help find these 4 numbers.



So $30 = 2 \times 3 \times 5$, these would be likely candidates to start with. Since there are only 3 numbers, 1 or -1 must be a factor. Also you can try 6, 10, 15, 30 and also the negative versions.

Try $(x + 1)$, re-write as $(x - -1)$

$$\begin{aligned}
 p(x) &= x^4 - 3x^3 - 15x^2 + 19x + 30 \\
 p(-1) &= (-1)^4 - 3 \times (-1)^3 - 15 \times (-1)^2 + 19 \times (-1) + 30 \\
 &= 1 + 3 - 15 - 19 + 30 \\
 &= 0
 \end{aligned}$$

Therefore $(x + 1)$ is a factor. Next divide $p(x)$ by $(x + 1)$

$$\begin{array}{r}
 \overline{x^3 - 4x^2 - 11x + 30} \\
 x+1 \overline{) x^4 - 3x^3 - 15x^2 + 19x + 30} \\
 \underline{x^4 + x^3} \\
 -4x^3 - 15x^2 \\
 \underline{-4x^3 - 4x^2} \\
 -11x^2 + 19x + 30 \\
 \underline{-11x^2 - 11x} \\
 30x + 30 \\
 \underline{30x + 30} \\
 0
 \end{array}$$

So $(x^3 - 4x^2 - 11x + 30)(x + 1)$

Now we can try $(x + 2)$, re-write as $(x - -2)$ and the quotient becomes the dividend

$$\begin{aligned}
 p(x) &= x^3 - 4x^2 - 11x + 30 \\
 p(-2) &= (-2)^3 - 4 \times (-2)^2 - 11 \times (-2) + 30 \\
 &= -8 - 16 + 22 + 30 \\
 &= 28
 \end{aligned}$$

Therefore $(x + 2)$ is not a factor

Try $(x-2)$,

$$p(x) = x^3 - 4x^2 - 11x + 30$$

$$\begin{aligned} p(2) &= (2)^3 - 4 \times (2)^2 - 11 \times (2) + 30 \\ &= 8 - 16 - 22 + 30 \\ &= 0 \end{aligned}$$

Therefore $(x-2)$ is a factor, now divide;

$$\begin{array}{r} x^2 - 2x - 15 \\ x-2 \overline{) x^3 - 4x^2 - 11x + 30} \\ \underline{x^3 - 2x^2} \\ -2x^2 - 11x \\ \underline{-2x^2 + 4x} \\ -15x + 30 \\ \underline{-15x + 30} \\ 0 \end{array}$$

So we now have $(x^2 - 2x - 15)(x-2)(x+1)$. You can factorise the quadratic using normal methods or continue the factor finding and division.

Since $-5 \times 3 = -15$ and $-5 + 3 = -2$, the last two factors will be $(x-5)$ and $(x+3)$. This leaves us with $(x+3)(x-5)(x-2)(x+1)$ as our final answer.

Exercises

15. Factorise $x^2 + 6x + 5$ using long division

Try $(x+5)$ $\therefore a = -5$

$$\begin{aligned} p(-5) &= (-5)^2 + 6(-5) + 5 \\ &= 25 - 30 + 5 \\ &= 0 \quad \therefore \text{factor.} \end{aligned}$$

$$\begin{array}{r} x+1 \\ x+5 \overline{) x^2 + 6x + 5} \\ \underline{x^2 + 5x} \\ x + 5 \\ \underline{x + 5} \\ 0 \end{array}$$

$$(x+5)(x+1)$$

16. Factorise $x^2 + 2x - 3$ using long division

Try $(x-3) \therefore a = 3$

$$p(x) = x^2 + 2x - 3$$

$$\begin{aligned} p(3) &= 3^2 + 2 \times 3 - 3 \\ &= 9 + 6 - 3 \\ &= 12 \therefore \text{not a factor} \end{aligned}$$

Try $(x+3) = (x-(-3)) \therefore a = -3$

$$\begin{aligned} p(-3) &= (-3)^2 + 2(-3) - 3 \\ &= 9 - 6 - 3 \\ &= 0 \therefore \text{factor} \end{aligned}$$

$$\begin{array}{r} x-1 \\ x+3 \overline{) x^2 + 2x - 3} \\ \underline{x^2 + 3x} \\ -x - 3 \\ \underline{-x - 3} \\ 0 \end{array}$$

$$(x+3)(x-1)$$

17. Factorise $x^3 + 2x^2 - 5x - 6$

$\begin{matrix} 6 \\ 2 \quad 3 \end{matrix}$ x^3 has 3 brackets, either $(x+1)$ or $(x-1)$ is a factor.

Try $(x+1) \therefore a = -1$

$$p(x) = x^3 + 2x^2 - 5x - 6$$

$$\begin{aligned} p(-1) &= (-1)^3 + 2(-1)^2 - 5(-1) - 6 \\ &= -1 + 2 + 5 - 6 \\ &= 0 \therefore \text{factor} \end{aligned}$$

$$\begin{array}{r} x^2 + x - 6 \\ x+1 \overline{) x^3 + 2x^2 - 5x - 6} \\ \underline{x^3 + x^2} \\ x^2 - 5x - 6 \\ \underline{x^2 + x} \\ -6x - 6 \\ \underline{-6x - 6} \\ 0 \end{array}$$

$$(x+1)(x^2 + x - 6)$$

$$(x+1)(x-2)(x+3)$$

18. Factorise $x^3 + 5x^2 - x - 5$

as 5 is prime and solution has 3 brackets $(x+1)$ and/or $(x-1)$ must be factors.

Try $(x-1) \therefore a=1$

$$P(x) = x^3 + 5x^2 - x - 5$$

$$\begin{aligned} P(1) &= 1^3 + 5 \times 1^2 - 1 - 5 \\ &= 1 + 5 - 1 - 5 \\ &= 0 \therefore \text{factor} \end{aligned}$$

$$\begin{array}{r} x^2 + 6x + 5 \\ x-1 \overline{) x^3 + 5x^2 - x - 5} \\ \underline{x^3 - x^2} \\ 6x^2 - x \\ \underline{6x^2 - 6x} \\ 5x - 5 \\ \underline{5x - 5} \\ 0 \end{array}$$

$$(x-1)(x^2 + 6x + 5)$$

$$(x-1)(x+1)(x+5)$$

19. Factorise $x^3 + x^2 - 4x - 4$

$$\begin{array}{c} 4 \\ \swarrow \searrow \\ 2 \quad 2 \end{array}$$

Try $(x-2) \therefore a=2$

$$P(x) = x^3 + x^2 - 4x - 4$$

$$\begin{aligned} P(2) &= 2^3 + 2^2 - 4 \times 2 - 4 \\ &= 8 + 4 - 8 - 4 \\ &= 0 \therefore \text{factor} \end{aligned}$$

$$\begin{array}{r} x^2 + 3x + 2 \\ x-2 \overline{) x^3 + x^2 - 4x - 4} \\ \underline{x^3 - 2x^2} \\ 3x^2 - 4x \\ \underline{3x^2 - 6x} \\ 2x - 4 \\ \underline{2x - 4} \\ 0 \end{array}$$

$$(x-2)(x^2 + 3x + 2)$$

$$(x-2)(x+2)(x+1)$$

End of Complete in-class Portion of the Investigation